ReportsNet: Rotation-Enhanced Polar-Radial Topology Synergy Network for Hyperspectral and LiDAR-Derived DSM Joint Classification

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***Abstract*—Joint classification of hyperspectral (HS) imagery and light detection and ranging (LiDAR)-derived digital surface models (DSMs) commonly represents local neighborhoods on Cartesian grids, where changes in object orientation alter the spatial arrangement presented to the fusion network. Existing fusion architectures do not explicitly encode center-relative radial distances, cyclic angular relationships, or geometrically corresponding cross-modal locations. To address this gap, the rotation-enhanced polar-radial topology synergy network (ReportsNet) is proposed, which locally reorganizes co-registered HS and DSM patches into target-centered polar-radial representations and models complementary distance-based and relative-angle-aware context for multimodal fusion. On the MUUFL, Trento, and Houston datasets, ReportsNet obtains OAs of 91.99%, 99.45%, and 90.69%, respectively. It achieves the highest OA on all three datasets, exceeding the next-best compared methods by 1.44, 0.44, and 1.28 percentage points, respectively. In the Houston rotation experiment from 0° to 90°, ReportsNet yields an OA standard deviation of 0.06%, indicating stable performance under the evaluated perturbations. The code is available at https://github.com/LuyangLyu/radial-flow**

***Index Terms*—Collaborative expert synergy, digital surface model (DSM), hyperspectral (HS), joint classification, polar-radial topology, rotation robustness.**

I. Introduction

REMOTE sensing techniques are widely used to characterize objects and spatial patterns on the Earth's surface [1], with applications including forest resource estimation, meteorological observation, and agricultural monitoring [2], [3]. Various types of satellites and aircraft have been applied to capture diverse RS modalities, including hyperspectral (HS) data [4], multispectral (MS) data [5], and light detection and ranging (LiDAR) data [6]. HS imagery characterizes surface materials through detailed reflectance spectra. However, illumination variations, cast shadows, atmospheric effects, and terrain slope or aspect can alter the measured reflectance of the same material, while different land-cover categories may exhibit overlapping spectral signatures, which reduces spectral class separability [7]. A LiDAR-derived digital surface model (DSM) does not directly correct these spectral distortions but provides complementary cues about surface elevation, relative height, and structural discontinuities. A DSM is a raster surface-elevation product rather than raw LiDAR returns or a full 3-D point cloud; although DSMs can also be generated by techniques such as structure-from-motion photogrammetry, the benchmark inputs used in this study are LiDAR-derived DSMs. Fusing the two modalities can therefore help distinguish categories that are spectrally similar but geometrically different and improve classification when HS signatures vary with environmental or topographic conditions. Existing studies have demonstrated the effectiveness of fusing HS imagery with LiDAR-derived information for joint land-cover classification [7], [8], [9], [10].

Existing HS-DSM joint classifiers mainly improve where or how heterogeneous features are fused. Separate encoders typically extract spectral–spatial cues from HS imagery and elevation-structure cues from the DSM before combining them in a shared representation. CNN-based methods strengthen local interaction through multibranch convolution [11], [12], [13], [14], [15], transformer-based methods model broader cross-modal dependencies [16], [17], [18], and Mamba-based methods reduce sequence-modeling complexity while retaining broad spatial context [19], [20]. These developments improve feature extraction and fusion but generally inherit Cartesian square patches: neighbors are indexed by horizontal and vertical offsets, and orientation changes alter the arrangement presented to the backbone. Consequently, stronger fusion modules alone do not explicitly encode the angular and center-to-periphery structure of a target-centered neighborhood.

A separate line of research addresses orientation changes through rotation-equivariant or rotation-invariant operators.

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Standard Cartesian convolution is translation equivariant, but its responses generally change when a local pattern is rotated unless rotation-specific constraints are introduced [21], [22]. Rotation augmentation can improve empirical robustness, but unlike explicitly equivariant operators, it does not impose an exact transformation law on learned features [23], [24]. Group-equivariant CNNs share filters over discrete transformation groups, harmonic networks constrain responses to transform predictably, Paoletti et al. [21] use circular harmonic filters for HSI, and RIC-CNN [22] encodes center-relative coordinates. These methods provide valuable geometric priors but mainly target single-image feature extraction or generic CNN replacement, rather than geometry-matched HS-DSM interaction.

Polar and log-polar representations offer another route by mapping an in-plane rotation about the selected center to a translation along the angular axis. Warped convolutions exploit this property for transformation-aware feature learning [25], while polar transformer networks learn a task-dependent transformation origin [26]. These studies demonstrate the value of polar-coordinate representations for handling spatial transformations, with a primary focus on single-image representation learning. For HS-DSM joint classification, the corresponding opportunity is to coordinate complementary spectral and elevation features at matched angular-radial locations while preserving the polar-radial organization during long-range interaction. Rotation-enhanced polar-radial topology synergy network (ReportsNet) realizes this extension through complementary ring-wise and radial-line modeling, topology-aware global interaction across radial lines, and geometry-aligned cross-modal attention.

The proposed representation uses polar-radial indexing as a target-centered organizational prior applicable to irregular roads, buildings, and vegetation. An ideal coordinate reparameterization does not create information or make polar-radial coordinates universally more expressive than Cartesian coordinates. Its role here is instead to expose two variables that are directly relevant to center-pixel classification: center-relative distance, which organizes contextual scale, and relative direction, which retains anisotropic transitions such as roads, boundaries, and elongated structures. Ring-wise modeling summarizes evidence at comparable distances, whereas radial-line modeling preserves direction-sensitive center-to-periphery structure; their combination provides a task-specific inductive bias rather than assuming that land-cover objects are circular.

Under ideal continuous sampling, $\mathcal{P}(\mathcal{R}_\theta X)(r,\phi) = \mathcal{P}(X)(r,\phi-\theta)$. Thus, a center-preserving in-plane rotation becomes an angular translation while the radius is preserved. ReportsNet instantiates this geometric relation through finite angular sampling, bilinear interpolation, and topology-aware learned operators, with the resulting rotation stability evaluated experimentally.

The evaluated perturbation is a joint in-plane rotation of a co-registered local HS-DSM neighborhood about its classified center pixel, representing changes in local land-cover orientation or patch presentation. Under such rotations, the polar-radial representation preserves radial coordinates and converts orientation changes into cyclic reordering along the angular dimension. ReportsNet combines angle-eliminated ring modeling, relative-angle-aware global interaction across radial lines, and geometry-aligned HS-DSM attention to maintain consistent classification. Rotation robustness is quantified through comparative accuracy variation over the evaluated angles.

The unresolved problem is therefore not simply to replace Cartesian coordinates or to impose generic rotation invariance, but to construct a target-centered representation that jointly preserves radial distance, angular ordering, and cross-modal correspondence. ReportsNet addresses this gap through local polar-radial resampling centered on each classified pixel, complementary ring and radial-line experts, and paired interaction between HS and LiDAR-derived DSM features at corresponding polar-radial locations.

Based on the above analysis, two related challenges remain in complex real-world scenarios.

1) *Limited-Sample Representation Learning:* Under class-wise training ratios as low as 0.5%–1%, each category has only a small labeled subset, making stable spectral-elevation learning difficult across illumination, topographic, orientation, and land-cover variations [27], [28], [29]. The limited-sample behavior is therefore evaluated at multiple ratios rather than inferred from model capacity or training accuracy alone.
2) *Orientation-Sensitive Multimodal Organization:* Cartesian fusion, rotation-aware convolution, and global polar canonicalization each address only part of the problem. Joint classification additionally requires HS and DSM context to remain paired by angle and distance when local orientations change.

To address these challenges, a ReportsNet is proposed for HS-DSM joint classification. Ring expert (RE) models angle-eliminated context at fixed radii, radial line expert (RLE) retains radial and relative-angular dependencies, and geometry-aligned interaction exchanges corresponding HS and DSM information between modalities. Controlled topology and rotation experiments quantify the rotation-enhanced behavior of the resulting classifier.

1) A target-centered polar-radial representation is introduced for HS-DSM joint classification to explicitly organize local evidence by center-relative distance and cyclic angular order, without assuming that land-cover objects have circular shapes.
2) A complementary representation strategy separates angle-eliminated distance evidence from relative-angle-aware radial evidence, models topology-aware global dependencies without discarding polar-radial organization, and enables HS-DSM interaction at corresponding neighborhood locations.
3) Experiments on three benchmarks, multiple label ratios, controlled in-plane rotations, and a spatially disjoint split provide evidence of competitive classification performance and stable behavior across the evaluated conditions. Capacity-matched controls further distinguish the benefit of the proposed representation from parameter count alone.

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Section II reviews related works. Section III details the proposed ReportsNet. Section V presents experimental results and analysis. Section VI concludes this article.

II. Related Works

A. HS and LiDAR Fusion and Joint Classification

Existing studies have verified the complementarity of HS imagery and LiDAR-derived information for joint classification [8], [9], [30]. Fusion strategies include feature-level combination of modality-specific representations and decision-level aggregation of parallel predictions, which may also be used together [16], [31], [32], [33], [34], [35]. HS supplies material-dependent spectral responses, although observed reflectance varies with illumination, atmosphere, and terrain-induced shading; the DSM supplies elevation and surface structure but may inherit LiDAR sensitivities to scan geometry, incidence angle, occlusion, and surface conditions [7], [27]. The two modalities thus contribute complementary measurements with distinct sensitivities. Brell et al. [27] jointly used HS reflectance and LiDAR intensity to correct illumination and anisotropic effects and support cross-calibration. This framework enhances radiometric consistency through multimodal calibration, while ReportsNet focuses on learned target-centered neighborhood organization. Subsequent methods introduced coupled CNN streams, shared-specific representations, cross-modal semantic enhancement, and local–global attention [13], [17], [18], [36], [37]. Although these architectures can learn circular, radial, and orientation-dependent patterns, their Cartesian neighborhoods encode center-relative distance, cyclic adjacency, and orientation variants only implicitly.

B. CNN- and Transformer-Based Techniques

Although discriminative and generative paradigms have both been explored in deep RS interpretation [38], [39], [40], supervised multimodal joint classification is primarily developed through CNN- and transformer-based feature learning. Representative discriminative designs strengthen local representation through spatial–spectral stacked CNNs with feedback attention and hybrid fully connected tensorized networks [41], [42]. For HS-DSM fusion, CNN architectures further employ multibranch, encoder–decoder, and attention modules to extract spectral–spatial and elevation-structure cues [13], [43]. Representative designs include interleaving perception CNN, shuffle feature enhancement, multiscale learning, mutual guidance attention, and self-guided multibranch fusion [14], [44], [45], [46], [47]. These modules improve local discriminability and cross-modal balance, while their Cartesian receptive fields retain orientation-dependent neighborhood organization.

Transformers complement local CNN modeling by capturing broader spatial dependencies, multiscale representations, and local–global collaboration [16], [17], [18], [37], [48], [49], [50], [51], [52]. Unsupervised transformer fusion has also been investigated for HS-MS data [53], demonstrating the broader applicability of attention-based multimodal interaction. For HS-DSM fusion, HI^{2}D^{2}FNet [28] decomposes HSI into reflectance and shading components, associates horizontal geometry in shading with vertical geometry from LiDAR-derived elevation, and uses the fused geometry feature to guide the reflectance stream. Although self-attention has quadratic complexity in token length, this cost remains manageable for small patches; the central distinction for the present work is that Cartesian tokenization organizes context by horizontal and vertical offsets instead of explicitly separating center-relative radius from cyclic angle.

C. Mamba-Based Methods

State-space models, particularly Mamba [54], provide linear-complexity long-range sequence modeling. In remote sensing, spatial–spectral Mamba for HS image classification (MambaHSI) [19] uses bidirectional state-space models to capture both spectral signatures and spatial distributions simultaneously. Subsequent cross-Mamba, multimodal fusion Mamba, and dynamic fusion strategies model complementary HS-DSM information and adaptively weight heterogeneous sensors [55], [56], [57]. These methods can learn orientation-dependent or circular patterns, but Cartesian serialization does not explicitly preserve cyclic angular adjacency and center-to-periphery ordering.

ReportsNet explicitly exposes center-relative distance, cyclic angular adjacency, and geometrically corresponding HS-DSM locations. This inductive bias reduces the need to learn orientation variants solely from Cartesian arrangements and supports complementary ring-wise, radial-line, and cross-modal modeling. Controlled topology, rotation, and label-ratio experiments quantify the contribution of this inductive bias.

D. Rotation-Aware Learning and Polar-Coordinate Representations

Group-equivariant and harmonic networks constrain feature transformations under rotation [23], [24]. For HSI, Paoletti et al. [21] replace conventional kernels with circular harmonic filters, whereas RIC-CNN [22] encodes positions relative to the input center and can replace conventional convolution without additional trainable parameters or rotation augmentation. Warped convolution and polar transformers use polar or log-polar representations for transformation-aware feature learning [25], [26]. These rotation-aware methods establish valuable geometric priors, with primary objectives in single-image representation learning or generic operator replacement. ReportsNet organizes paired HS-DSM observations into corresponding ring and radial-line tokens. Radial-angular mixer (RAM) combines circular local modeling with relative-angle-conditioned global interaction across all radial lines, while paired cross-modal attention (PCMA) exchanges cross-modal information at matched polar-radial locations. Controlled rotation experiments quantify the stability achieved by this topology-aware multimodal design.

III. Methodology

A. Overview

The detailed architecture of ReportsNet is illustrated in Fig. 1. As shown in Fig. 2, ReportsNet locally reorganizes each

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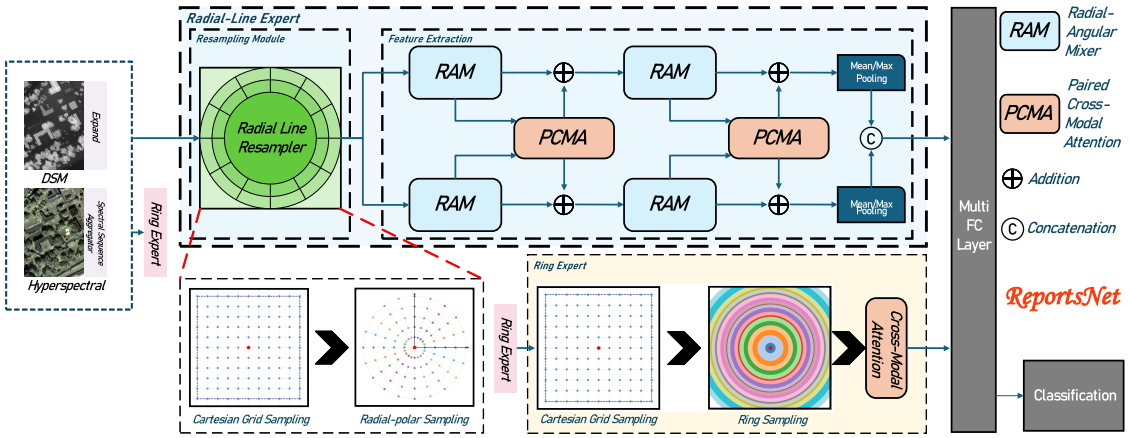


Fig. 1. Scheme of ReportsNet for HS and LiDAR-derived DSM joint classification.

target-centered Cartesian feature patch into ring-wise tokens and bilinearly sampled radial-line tokens while retaining the co-registered HS and DSM images on their original grids. The network integrates a complementary RE and RLE for this local representation. RE eliminates the angular index through permutation-invariant aggregation within each ring and retains the center-relative distance level, whereas RLE retains radial positions and cyclic relative angular relationships without encoding an absolute angular origin. Accordingly, RE provides a distance-only representation, while RLE provides a joint relative radius-angle representation through stacked RAM blocks with PCMA. Each RAM block integrates circular polar convolution with hierarchical polar rotary position (HPRP) encoding, as illustrated in Fig. 3. Cross-modal interaction is implemented through bidirectional cross-attention and gated fusion over ring tokens in RE and through PCMA over geometrically corresponding radial-line tokens in RLE.

1) Representation Rationale: The motivation for polar-radial organization is a structured factorization of target-centered context rather than an intrinsic information advantage over a Cartesian grid. Under ideal continuous resampling, let $\mathbf{P}_\star(r,\phi)$ denote the feature of modality $\star \in \{H, L\}$ at center-relative distance r and direction ϕ. It can be decomposed into an angular mean and an a-direction-dependent residual as

$$\mathbf{P}_\star(r,\phi) = \boldsymbol{\mu}_\star(r) + \boldsymbol{\delta}_\star(r,\phi) \tag{1}$$

$$\boldsymbol{\mu}_\star(r) = \frac{1}{2\pi}\int_0^{2\pi} \mathbf{P}_\star(r,\varphi)\, d\varphi \tag{2}$$

$$\int_0^{2\pi} \boldsymbol{\delta}_\star(r,\phi)\, d\phi = \mathbf{0}. \tag{3}$$

The radial term $\boldsymbol{\mu}_\star(r)$ describes how the average context changes with distance from the classified pixel, whereas $\boldsymbol{\delta}_\star(r,\phi)$ retains anisotropic departures from that distance-level context. RE discretely models the first type of evidence through permutation-invariant mean and max statistics within equal-distance rings, with max pooling additionally retaining salient responses that angular averaging may suppress. RLE preserves the ordered joint (r,ϕ) structure needed to model the second type, including boundaries and elongated or curved patterns whose direction changes with distance. In principle, a Cartesian network can learn the same relations, but they are implicit in horizontal-vertical offsets; the proposed organization exposes them directly and permits shared processing across comparable distances and relative directions. Thus, the theoretical justification concerns the inductive bias and complementary factorization used by the complete model, not a claim that coordinate conversion alone increases representational information.

B. Spectral Projection and Sequence Aggregation

The inputs are an HS patch $\mathbf{X}_H \in \mathbb{R}^{C_{in}\times P\times P}$ and a co-registered LiDAR-derived DSM patch $\mathbf{X}_L \in \mathbb{R}^{C_L\times P\times P}$, where C_{in} and C_L denote their channel numbers and P is the patch size. The subscript L compactly identifies this DSM branch throughout the formulation. For the HS input, a 1×1 convolution layer is applied to project the spectral dimension C_{in} to a fixed hidden dimension C. For the single-channel DSM ($C_L = 1$), Sobel operators are applied in the horizontal and vertical directions to obtain the local elevation-gradient magnitude. The gradient magnitude is concatenated with the original DSM feature to form $\mathbf{F}_L$ as follows:

$$\begin{aligned}\mathbf{G}_L &= \sqrt{(\mathbf{S}_x * \mathbf{X}_L)^2 + (\mathbf{S}_y * \mathbf{X}_L)^2 + \epsilon} \\ \mathbf{F}_L &= \left[\mathbf{X}_L \,\middle\|\, \mathbf{G}_L\right]\end{aligned} \tag{4}$$

where $\mathbf{S}_x$ and $\mathbf{S}_y$ are the Sobel kernels, $\|$ denotes channel-wise concatenation, and ϵ is a small constant. The Sobel magnitude emphasizes local slopes and surface-height discontinuities in the DSM. This fixed preprocessing provides a low-cost explicit gradient prior when labeled samples are limited. The S_{14} DSM-without-Sobel ablation in Table I leaves the parameter count unchanged and, in the paired preprocessing comparison, reduces OA and AA by 0.35 and 2.80 percentage points, respectively. Lightweight projections in the subsequent expert modules align the DSM feature dimension to C. Both obtained features are processed by layer normalization and activated by the GELU function.

After spectral projection, the features are processed by a spectral sequence aggregator (SSA) to capture long-range spectral dependencies. Denoting the projected HS features

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as $\mathbf{F}_{in} \in \mathbb{R}^{B\times C\times P\times P}$, the SSA adopts a parallel architecture that integrates convolutional local feature extraction with bidirectional state-space modeling. Specifically, the SSA consists of two parallel branches. The first branch is a convolutional branch that applies 1×1 point-wise convolution to capture local spectral correlations. The second branch is a bidirectional Mamba branch designed to model long-range sequential dependencies. The Mamba branch partitions the C channels into G groups of $\lceil C/G \rceil$ channels each, forming a sequence $\mathbf{F}_{seq} \in \mathbb{R}^{(BP^2)\times G\times \lceil C/G \rceil}$ that is processed in both forward and backward directions

$$\begin{aligned}\mathbf{F}_{conv} &= \mathrm{Conv}_{1\times 1}(\mathbf{F}_{in})\\ \mathbf{F}_{mamba} &= \mathrm{Mamba}_{forward}(\mathbf{F}_{seq})\\ &\quad + \mathrm{Mamba}_{backward}\left(\mathbf{F}_{seq}^{rev}\right)^{rev}\end{aligned} \tag{5}$$

where $(\cdot)^{rev}$ denotes the reversal operation. Both branch outputs are aggregated by element-wise addition, followed by layer normalization and GELU activation, as follows:

$$\mathbf{F}_H = \mathbf{F}_{in} + \mathrm{GELU}\left(\mathrm{LN}\left(\mathbf{F}_{conv} + \mathbf{F}_{mamba}\right)\right). \tag{6}$$

This parallel design allows the SSA to simultaneously capture local spectral patterns through convolution and long-range dependencies through state-space modeling, with a residual connection preserving the original feature information. The processed HS features $\mathbf{F}_H$ are subsequently fed into two parallel experts as follows:

$$\mathbf{f}_{RE} = \mathrm{RE}(\mathbf{F}_H, \mathbf{F}_L), \quad \mathbf{f}_{RLE} = \mathrm{RLE}(\mathbf{F}_H, \mathbf{F}_L). \tag{7}$$

C. Ring Expert

RE summarizes multimodal context at comparable distances from the classified center pixel. It groups HS and DSM features into N_r concentric rings according to Euclidean distance from the patch center. Because mean and max pooling are invariant to permutations of the samples within a ring, the ring token discards their angular identities and ordering while retaining the ring's center-relative distance level. RE therefore forms a dedicated distance-only path for learning how contextual evidence at different radii contributes to the center-pixel prediction. The contribution of this discrete distance-only representation is quantified through controlled topology and rotation experiments.

For a patch of size $P \times P$, all pixels are grouped into N_r concentric rings based on their Euclidean distance from the center. The input feature map is denoted as $\mathbf{F}_\star \in \mathbb{R}^{B\times C\times P\times P}$; its batch dimension B is omitted hereafter for clarity; the subscript $\star$ denotes operations applied identically to both modalities, that is, $\star \in \{H, L\}$, and $\bar{\star}$ denotes the complementary modality. The ring representation is formulated by computing mean and max pooling statistics for each ring as follows:

$$\mathbf{r}_{\star,i}^{mean} = \frac{1}{|R_i|}\sum_{(x,y)\in R_i}\mathbf{F}_\star(x,y), \quad \mathbf{r}_{\star,i}^{max} = \max_{(x,y)\in R_i}\mathbf{F}_\star(x,y) \tag{8}$$

where R_i denotes the set of pixel coordinates in ring i, and $\mathbf{F}(x,y) \in \mathbb{R}^C$ denotes the feature vector at spatial location (x,y). The ring representation is formulated by concatenating the mean and max statistics as follows:

$$\mathbf{r}_{\star,i} = \left[\mathbf{r}_{\star,i}^{mean} \middle\| \mathbf{r}_{\star,i}^{max}\right] \in \mathbb{R}^{2C} \tag{9}$$

where C is the feature channel dimension. At the origin, $R_1 = \{(x_c, y_c)\}$ is a singleton ring with radius zero; hence, both its mean and max statistics reproduce the center-pixel feature without mixing it with surrounding samples. The subsequent inner rings represent nearby pixels grouped by their discrete Euclidean distances from the center. For the DSM branch, the preprocessed feature $\mathbf{F}_L$ in Section III-B is processed directly by radial ring pooling. The pooled HS and DSM features are fused using dual-stream cross-attention. By stacking all ring representations, the ring feature matrix $\mathbf{R}_\star = [\mathbf{r}_{\star,1}; \mathbf{r}_{\star,2}; \cdots; \mathbf{r}_{\star,N_r}] \in \mathbb{R}^{N_r\times 2C}$ can be obtained. Given the HS ring feature $\mathbf{R}_H$ and DSM ring feature $\mathbf{R}_L$, lightweight projections first map them to a shared hidden dimension, after which bidirectional cross-attention and gated fusion are applied as follows:

$$\begin{cases}\mathbf{R}'_\star = \mathrm{LN}\left(\mathrm{MHA}\left(\mathbf{R}_\star, \mathbf{R}_{\bar{\star}}, \mathbf{R}_{\bar{\star}}\right) + \mathbf{R}_\star\right)\\ \mathbf{g} = \sigma\left(\mathbf{W}_g\left[\mathbf{R}'_H \| \mathbf{R}'_L\right] + \mathbf{b}_g\right)\\ \mathbf{R}_{fused} = \mathbf{g}\odot\mathbf{R}'_H + (1-\mathbf{g})\odot\mathbf{R}'_L\end{cases} \tag{10}$$

where MHA$(\cdot)$ denotes the multihead attention operation, and LN$(\cdot)$ denotes layer normalization. This dual-stream design enables effective bidirectional information exchange while preserving modality-specific features. Denoting the ith ring token of $\mathbf{R}_{fused}$ as $\mathbf{r}_i^{RE} \in \mathbb{R}^{2C}$, the pooled RE output for synergy fusion can be formulated as follows:

$$\mathbf{f}_{RE}^{pooled} = \frac{1}{N_r}\sum_{i=1}^{N_r}\mathbf{r}_i^{RE} \in \mathbb{R}^{2C}. \tag{11}$$

D. Radial Line Expert

Unlike the distance-only RE, RLE preserves both the ordered radial positions along each line and the cyclic relative ordering among lines. At the discrete topology and operator-design level, RLE represents angles relationally: circular padding treats the angular axis cyclically, HPRP uses pairwise angular offsets, and radial positional encoding is shared across angular directions. Specifically, RLE processes features sampled along radial lines and applies iterative refinement through stacked RAM blocks with PCMA. RLE captures angular structures along K discrete angular directions uniformly distributed in $[0, \pi)$, where K is a dataset-specific hyperparameter tuned by validation. For each angle θ_k, L points are sampled at regular intervals along a diameter passing through the center, transforming the input from $\mathbf{F}_\star \in \mathbb{R}^{C\times P\times P}$ to $\mathbf{L}_\star^{raw} \in \mathbb{R}^{K\times L\times C}$, as follows:

$$\begin{aligned}\mathbf{l}_k = \{&\mathbf{F}_\star(x_c + d\cos\theta_k, y_c + d\sin\theta_k) \mid\\ &d = -\lfloor L/2\rfloor, \ldots, \lfloor L/2\rfloor\}\end{aligned} \tag{12}$$

where (x_c, y_c) denotes the patch center. Because L is odd, the middle radial index has $d = 0$ and samples exactly (x_c, y_c) on every radial line. At $d = 0$, the three subdirections coincide at the same integer center coordinate, so bilinear sampling

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returns the exact center feature, and their average leaves it unchanged. The angular coordinate is undefined at this origin, so the shared center sample uses a common representation and radial positional encoding across all angular directions. Consequently, the center pixel is retained exactly as the shared middle sample of every radial line and participates in the subsequent RLE processing and pooling. For each nominal direction θ_k, the implementation additionally samples the three subdirections $\theta_k - \pi/(4K)$, θ_k, and $\theta_k + \pi/(4K)$ and averages their features, reducing discretization sensitivity for locations near the origin. Since the sampled coordinates (x, y) are generally noninteger positions, bilinear interpolation is employed to obtain feature values. For a continuous coordinate (x, y), the surrounding integer grid points are denoted as $\{x_0, x_1, y_0, y_1\}$, where $x_0 = \lfloor x \rfloor$, $x_1 = x_0 + 1$, $y_0 = \lfloor y \rfloor$, $y_1 = y_0 + 1$, $\lfloor \cdot \rfloor$ denotes the floor operation. $\alpha = x - x_0$ and $\beta = y - y_0$ are the fractional offsets. The bilinearly interpolated feature can be obtained as follows:

$$\begin{aligned}\mathbf{F}(x, y) = (1-\alpha)(1-\beta)\mathbf{F}_{00} + \alpha(1-\beta)\mathbf{F}_{10} \\ + (1-\alpha)\beta\mathbf{F}_{01} + \alpha\beta\mathbf{F}_{11}\end{aligned} \tag{13}$$

where $\mathbf{F}_{ij} = \mathbf{F}(x_i, y_j)$ denotes the features at the four neighboring grid points. The same sampling is applied to both modalities (i.e., $\mathbf{F}_\star \in \{\mathbf{F}_H, \mathbf{F}_L\}$). The implementation performs bilinear sampling on normalized coordinates within $[-1, 1]^2$. Under ideal continuous sampling, an in-plane rotation about the patch center corresponds to a cyclic shift of the sampled angular directions while preserving radial coordinates. Let $\Delta\phi = 2\pi/(2K)$ denote the angular interval after mirroring. When the rotation angle satisfies $\theta = m\Delta\phi$ for an integer m, and interpolation and boundary effects are ignored, the resampled $2K \times L$ polar-radial matrix undergoes only a circular shift of m rows along its angular dimension. For arbitrary angles, finite angular sampling and bilinear interpolation introduce approximation errors, so the resulting stability is quantified empirically rather than treated as exact continuous-rotation invariance. Conversely, for rotations that are exact symmetries of the square pixel grid (integer multiples of 90°), the lattice maps onto itself without resampling, so the interpolation and boundary effects vanish and the angular grid undergoes an exact cyclic shift; the complete network is therefore exactly invariant under these rotations, with the approximation above confined to nonaxis-aligned angles.

Fig. 2 visualizes the actual local resampling applied to one 11×11 patch. The Cartesian neighborhood is reorganized into a $2K \times L$ grid whose rows index relative angular directions and whose columns retain ordered radial positions, with the dashed middle column indicating the shared $d = 0$ center sample.

To enhance angular coverage and enable bidirectional angular attention, angular mirroring is applied by concatenating each radial line with its flipped version, expanding the feature representation from K lines to $2K$ lines. This design ensures complete angular coverage as follows: the original K lines cover angles $[0, \pi)$, while the flipped K lines cover angles $[\pi, 2\pi)$. Learnable radial positional encodings are subsequently added as follows:

$$\mathbf{L}_\star^{pe} = \left[\mathbf{L}_\star^{raw} \| \text{Flip}\left(\mathbf{L}_\star^{raw}\right)\right] + \mathbf{E}_{rad} \tag{14}$$

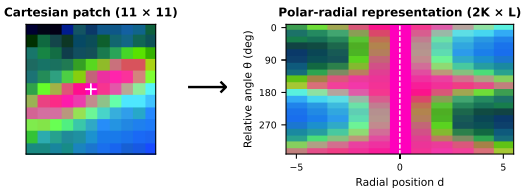


Fig. 2. Visualization of a target-centered Cartesian patch and its locally resampled polar-radial representation.

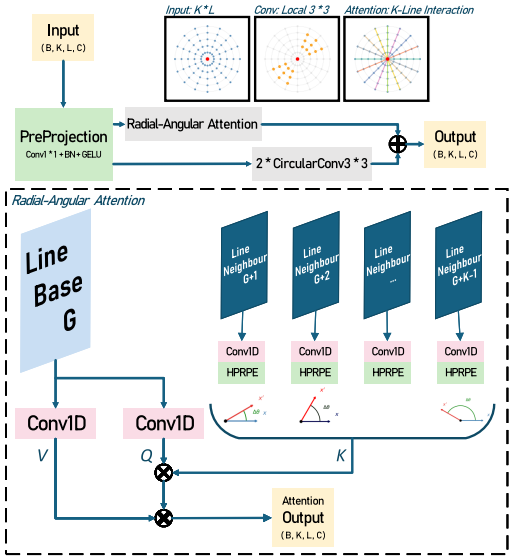


Fig. 3. Detailed structure of the RAM block for angular dependency modeling.

where $\mathbf{E}_{rad} \in \mathbb{R}^{1\times L\times C}$ encodes the radial distance from the patch center and is shared across all angular directions. Because $\mathbf{E}_{rad}$ depends on radial position L instead of angular index K, it remains unchanged under cyclic reordering of the angular samples and preserves a common radial reference across orientations. In (14), $\mathbf{L}_\star^{raw} \in \mathbb{R}^{K\times L\times C}$ denotes the original K radial line features, and $\mathbf{L}_\star^{pe} \in \mathbb{R}^{2K\times L\times C}$ is the final feature after mirroring and radial positional encoding.

As the fundamental building unit of RLE, RAM takes $\mathbf{Z} = \mathbf{L}_\star^{pe} \in \mathbb{R}^{B\times 2K\times L\times C}$ and applies a local circular-convolution branch and a relative-angle-conditioned attention branch in parallel after a shared pointwise projection, as illustrated in Fig. 3. Their outputs are combined with the block input through a residual connection.

1) Local Branch With Circular Polar Convolution: Circular convolution with a kernel size of 3 and circular padding along the angular dimension captures cyclic-shift-aware local patterns on the sampled polar grid as follows:

$$\mathbf{Z}_{local} = \text{CPC2D}_{3\times 3}(\text{LN}(\mathbf{Z})) \tag{15}$$

where CPC2D denotes a circular polar convolution applied on the 2-D grid formed by the angular dimension $(2K)$ and the radial dimension (L) of $\mathbf{Z}$. Specifically, $\text{CPC2D}_{3\times 3}$ performs a 3×3 convolution over the (angular, radial) axes with circular padding along the angular axis and standard padding along the radial axis. The circular padding ensures seamless 0°–360° continuity after angular mirroring.

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2) Radial-Angular Attention Branch: The radial-angular attention branch uses HPRP to aggregate relative-angle-conditioned keys across angular lines and then models radial dependencies within each line. This branch uses a single attention head, while the cross-modal interaction modules use four heads. Point-wise 1-D convolutions are used to project the query, key, and value features.

Different from additive positional biasing, HPRP rotates the key features directly in channel pairs. For each channel pair $(2c, 2c+1)$ at radial position l and angular offset $\theta_{ij} = 2\pi((j-i) \bmod 2K)/(2K)$ between query line i and source line j, the rotated key can be calculated as follows:

$$\begin{bmatrix} \widehat{\mathbf{K}}_{i,j}^{(2c,l)} \\ \widehat{\mathbf{K}}_{i,j}^{(2c+1,l)} \end{bmatrix} = \begin{bmatrix} \cos\phi_{c,l,ij} & -\sin\phi_{c,l,ij} \\ \sin\phi_{c,l,ij} & \cos\phi_{c,l,ij} \end{bmatrix} \begin{bmatrix} \mathbf{K}_j^{(2c,l)} \\ \mathbf{K}_j^{(2c+1,l)} \end{bmatrix} \tag{16}$$

where $\phi_{c,l,ij} = l\omega_c + \theta_{ij}$ is the combined rotation angle, $\omega_c = \beta_0^{-c/D}$, $D = C/2$, and $\beta_0 = 10\,000$. This formulation combines radial position l with the relative angular offset θ_{ij}; no absolute angle value for line i or j is supplied to HPRP.

Since the implementation constructs rotary keys for all relative angular offsets, these angle-conditioned keys are further aggregated to obtain the final key used by line i as follows:

$$\bar{\mathbf{K}}_i = \frac{1}{2K} \sum_{j=1}^{2K} \widehat{\mathbf{K}}_{i,j}. \tag{17}$$

The resulting keys are used to compute dependencies along the radial sequence of each line as follows:

$$\mathbf{A}_i = \mathrm{Softmax}\left(\frac{\mathbf{Q}_i \bar{\mathbf{K}}_i^T}{\sqrt{L}}\right), \quad \mathbf{Z}_{angular,i} = \mathbf{A}_i \mathbf{V}_i. \tag{18}$$

Here, $\sqrt{L}$ is the fixed attention temperature used in the implementation. The attention output is finally projected back to the channel space and added to the input through a residual connection structure. Together with the local circular convolution branch, the block preserves the "local-angular" decomposition for effective spatial feature extraction.

RLE also incorporates PCMA for bidirectional exchange between HS and DSM features on corresponding angular lines. Given the RAM outputs $\mathbf{U}_H^{(t)}, \mathbf{U}_L^{(t)} \in \mathbb{R}^{B\times 2K\times L\times C}$, PCMA is applied independently to each angular line over its L radial tokens as follows:

$$\begin{aligned} \mathbf{L}_H^{(t)} &= \mathrm{LN}\Big(\mathbf{U}_H^{(t)} + \mathrm{MHA}\left(\mathbf{U}_H^{(t)}, \mathbf{U}_L^{(t)}, \mathbf{U}_L^{(t)}\right)\Big) \\ \mathbf{L}_L^{(t)} &= \mathrm{LN}\Big(\mathbf{U}_L^{(t)} + \mathrm{MHA}\left(\mathbf{U}_L^{(t)}, \mathbf{U}_H^{(t)}, \mathbf{U}_H^{(t)}\right)\Big). \end{aligned} \tag{19}$$

Here, $\mathrm{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ denotes multihead attention, so each modality queries the other while retaining its own feature through the residual connection. RLE stacks T RAM-PCMA stages for iterative refinement, where T is a dataset-specific hyperparameter tuned by validation. After the final layer, global average and max statistics are extracted from both modalities as follows:

$$\begin{aligned} \mathbf{f}_{RLE}^{pooled} = \Big[&\mathrm{Mean}\left(\mathbf{L}_H^{(T)}\right) \| \mathrm{Max}\left(\mathbf{L}_H^{(T)}\right) \\ &\| \mathrm{Mean}\left(\mathbf{L}_L^{(T)}\right) \| \mathrm{Max}\left(\mathbf{L}_L^{(T)}\right)\Big]. \end{aligned} \tag{20}$$

Here, Mean$(\cdot)$ and Max$(\cdot)$ denote global pooling over the angular and radial dimensions (i.e., over $2K$ lines and L sampled positions), producing $\mathbb{R}^C$ vectors. Under the ideal bin-aligned condition above, RE remains unchanged because its within-ring mean and max aggregation is insensitive to the corresponding sample reordering. In RLE, the shared radial encoding, circular angular operations, relative-angle interaction, and line-wise PCMA map the circularly shifted input to a correspondingly shifted output, since the co-registered HS and DSM lines shift together. The final order-independent mean and max pooling removes this angular shift; consequently, the pooled synergy representation and predicted class remain unchanged under this idealized condition.

E. Collaborative Expert Synergy

The outputs from RE and RLE are fused through a collaborative synergy mechanism because they retain complementary geometric information rather than duplicate the same polar representation. The RE output provides angle-eliminated distance-level context, whereas the RLE output provides joint radial and relative-angular structure. For a coupled ring-radial pattern such as a spiral-like road, RE records the persistence of the corresponding evidence over successive center-relative distance levels but intentionally discards its angular trajectory. In the RLE polar grid, the same pattern appears as a feature response whose angular position changes with radial position. RAM models this coupling through three coordinated operations: circular polar convolution captures local changes across adjacent radial positions and cyclically adjacent lines, HPRP encodes radial position together with pairwise relative angular offsets, and the HPRP-conditioned line-interaction attention aggregates relative-angle-conditioned keys from all lines before modeling dependencies along each radial sequence. Thus, RLE retains the relative angular trajectory required for the coupled pattern, while RE supplies complementary angle-eliminated distance context for the final synergy representation. The innermost three RE rings and angular-order-pooled statistics from the RLE center region are concatenated to form center-region features. Let $\mathbf{r}_i^{RE} \in \mathbb{R}^{2C}$ denote the fused token of the ith ring produced by RE, as follows:

$$\begin{aligned} \mathbf{f}_{center} = \Big[&\mathbf{r}_1^{RE} \| \mathbf{r}_2^{RE} \| \mathbf{r}_3^{RE} \\ &\| \boldsymbol{\mu}_H^{center} \| \boldsymbol{\mu}_L^{center} \| \mathbf{m}_H^{center} \| \mathbf{m}_L^{center}\Big]. \end{aligned} \tag{21}$$

Here, $\boldsymbol{\mu}_H^{center}, \boldsymbol{\mu}_L^{center} \in \mathbb{R}^C$, and $\mathbf{m}_H^{center}, \mathbf{m}_L^{center} \in \mathbb{R}^C$ are mean and max statistics computed over all $2K$ angular samples after mirroring and over the three central radial indices $\mathcal{S}_c = \{\lfloor L/2 \rfloor - 1, \lfloor L/2 \rfloor, \lfloor L/2 \rfloor + 1\}$. These pooled statistics remain unchanged under an exact cyclic permutation of the sampled angular index and provide angular-order-independent center supervision. The center feature is processed through a center head CenterHead$(\cdot)$, implemented as a lightweight MLP classifier, as follows:

$$\hat{\mathbf{y}}_{center} = \mathrm{CenterHead}\left(\mathbf{f}_{center}\right). \tag{22}$$

Here, $\hat{\mathbf{y}}_{center}$ denotes the class logits for auxiliary center supervision. Similar to depth-wise deep supervision, CenterHead$(\cdot)$

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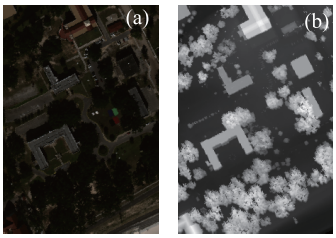


Fig. 4. Visualization of the MUUFL dataset, including (a) HS false-color image and (b) LiDAR-derived DSM gray image.

is used only during training to provide additional gradients on the angular-order-pooled center features and is discarded during inference. The final synergy feature combines statistical information from both experts as follows:

$$\mathbf{f}_{syn} = \left[\mathbf{f}_{RLE}^{pooled} \,\middle\|\, \mathbf{f}_{RE}^{pooled}\right] \tag{23}$$

where $\mathbf{f}_{RLE}^{pooled}$ and $\mathbf{f}_{RE}^{pooled}$ denote the pooled features from RLE and RE, respectively. This combined feature is processed through a multilayer perceptron as follows:

$$\hat{\mathbf{y}}_{syn} = \mathbf{W}_2\left(\mathrm{GELU}\left(\mathrm{LN}\left(\mathbf{W}_1\mathbf{f}_{syn} + \mathbf{b}_1\right)\right)\right) + \mathbf{b}_2. \tag{24}$$

The training objective combines cross-entropy losses for both main and auxiliary predictions as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{main} + w_{depth} \cdot \mathcal{L}_{depth} + w_{center} \cdot \mathcal{L}_{center}. \tag{25}$$

Here, $\mathcal{L}_{main} = \mathrm{CE}(\hat{\mathbf{y}}_{syn}, \mathbf{y})$ is the main classification loss on the synergy output, $\mathcal{L}_{center} = \mathrm{CE}(\hat{\mathbf{y}}_{center}, \mathbf{y})$ supervises the center-region prediction head, and $\mathcal{L}_{depth}$ provides deep supervision at the RLE stages. Auxiliary classifiers attached to all T stages produce logits $\hat{\mathbf{y}}_{depth}^{(t)}$, and $\mathcal{L}_{depth} = \sum_{t=1}^{T} \mathrm{CE}(\hat{\mathbf{y}}_{depth}^{(t)}, \mathbf{y})$, where $\mathbf{y}$ denotes the class label and w_{depth} and w_{center} are dataset-specific weights.

IV. Experimental Setup

A. Datasets and Data Partitions

The MUUFL dataset was captured over the University of Southern Mississippi Gulf Park Campus, USA. The HS data was acquired by the CASI-1500 sensor, containing 64 spectral channels ranging from 375 to 1050 nm. The public benchmark supplies a preprocessed single-channel surface-elevation raster derived from LiDAR point-cloud measurements and co-registered with the HS imagery [58]. This provider-supplied DSM is directly used; the registered HS image and DSM both have dimensions of 325 × 220 pixels, with spatial sampling of approximately 0.54 m across track and 1.0 m along track [58]. The dataset contains 11 land-cover categories. The detailed sample distribution is provided in Table I, and the visualization is shown in Fig. 4.

The Trento dataset was acquired over the south of Trento, Italy. The HS data were acquired by the AISA Eagle sensor, containing 63 spectral channels ranging from 402.89 to 989.09 nm. The LiDAR measurements were acquired with an Optech ALTM 3100EA sensor, and the co-registered LiDAR-derived DSM is used as the elevation input. The dataset has a spatial resolution of 1 m and an image size of 166 × 600 pixels.

TABLE I
Component Ablations and Model Capacity on Houston

Variant	Params. (M)	OA	ΔOA	AA	F1
S_1	0.910	89.93	−0.76	86.00	86.22
S_2	0.681	88.80	−1.89	84.73	84.63
S_3	0.977	90.12	−0.57	86.77	86.57
S_4	0.977	89.73	−0.96	86.83	86.77
S_5	0.977	89.55	−1.14	86.12	86.29
S_6	0.910	89.22	−1.47	85.79	85.71
S_7	0.812	89.54	−1.15	84.52	85.26
S_8	0.812	89.82	−0.87	86.56	86.27
S_9	0.495	88.43	−2.26	84.28	84.46
S_{10}	0.912	89.96	−0.73	85.90	86.44
S_{11}	1.195	88.51	−2.18	84.21	84.22
S_{12}	0.999	88.49	−2.20	83.12	84.06
S_{13}	0.977	90.32	−0.37	84.69	85.88
S_{14}	0.977	90.34	−0.35	85.05	85.95
Ours	0.977	**90.69**	0.00	**87.85**	**88.04**

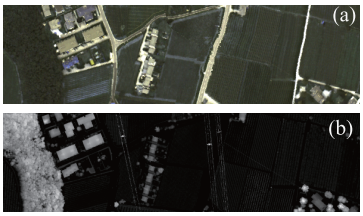


Fig. 5. Visualization of the Trento dataset, including (a) HS false-color image and (b) LiDAR-derived DSM gray image.

TABLE II
Training and Testing Sample Numbers in the Trento Dataset

Class	Land-cover Category	Training	Testing	Total
1	Apple Trees	40	3994	4034
2	Buildings	29	2874	2903
3	Ground	5	474	479
4	Wood	91	9032	9123
5	Vineyard	105	10396	10501
6	Roads	32	3142	3174

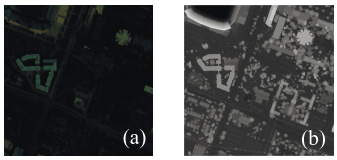


Fig. 6. Visualization of the Houston dataset, including (a) HS false-color image and (b) LiDAR-derived DSM gray image.

The dataset contains six land-cover categories. The detailed sample distribution is provided in Table II, and the visualization is shown in Fig. 5.

The Houston dataset was acquired over the University of Houston campus by the National Center for Airborne Laser Mapping (NCALM). The HS data has 48 spectral channels ranging from 380 to 1050 nm. The spatial resolutions of the HS imagery and the LiDAR-derived DSM used in this study are 1 m and 50 cm, respectively. The DSM is resampled to the HS grid before co-registered 11 × 11 patches are extracted from both modalities. A subscene (UH_NAD83_272652_3289689) with dimensions 601 × 596 was selected for experiments. The dataset contains

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TABLE III
TRAINING AND TESTING SAMPLE NUMBERS IN THE HOUSTON DATASET

Class	Land-cover Category	Training	Testing	Total
1	Healthy Grass	14	1439	1453
2	Stressed Grass	51	5020	5071
3	Evergreen Trees	64	6371	6435
4	Deciduous Trees	30	2925	2955
5	Residential Buildings	81	7982	8063
6	Non-Residential Buildings	482	47782	48264
7	Roads	70	6969	7039
8	Sidewalks	125	12372	12497
9	Major Thoroughfares	179	17722	17901
10	Paved Parking Lots	42	4114	4156
11	Cars	11	1090	1101
12	Stadium Seats	68	6756	6824

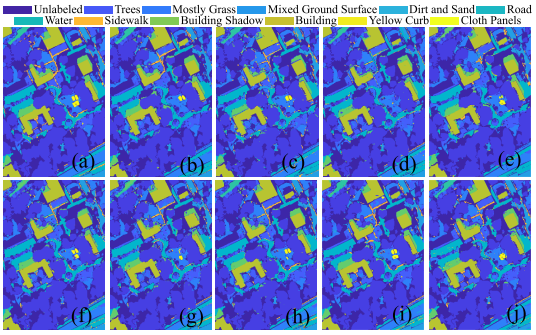


Fig. 7. (a) Ground truth and classification maps on the MUUFL dataset, including (b) MSNetSC, (c) LIIT, (d) HDML, (e) ExViT, (f) MambaHSI, (g) GMamba, (h) HugIpuNet, (i) CHNet, and (j) Ours.

12 land-cover categories. The detailed sample distribution is provided in Table III, and the visualization is shown in Fig. 6.

After excluding categories below 2‰ of all labels, 1% of each retained class is randomly selected for training and 99% for testing, following a recent limited-label HS-DSM protocol [29]. All methods used the same class-wise sampling protocol without class-frequency weighting or resampling. The main experiments are repeated ten times and report OA, AA, macro-PR, and macro-$F1$; multiple label ratios are evaluated separately. Because each class contributes equally to AA, macro-PR, and macro-$F1$, these measures complement OA when some categories contain very few training samples.

B. Implementation Settings and Compared Methods

The experiments were conducted using PyTorch 2.3 on two NVIDIA RTX 2080 Ti GPUs. The Adam optimizer was used with a learning rate of 0.0001 and weight decay of 0.01. A cosine annealing scheduler with warmup was employed for learning rate scheduling. The batch size was set to 16. The number of training epochs was set to 150 for Trento and 200 for the MUUFL and Houston datasets. The default configuration uses a patch size of 11×11 and a hidden dimension $C = 64$. The validation-selected settings are $w_{depth} = 0.1$, $w_{center} = 0.1/0.2/0.1$, and $(K, T) = (10, 3)/(10, 5)/(10, 2)$ for MUUFL/Trento/Houston; random horizontal and vertical flips are used. The comparison covers pixel-wise, single-modal, CNN-, transformer-, and Mamba-based methods [7], [15], [18], [19], [20], [29], [34], [43], [52], [58], [59], together with RIC-CNN [22] and rotation equivariant CNN (RotEq-CNN) [21] under the same 1% protocol. MambaHSI denotes spatial–spectral Mamba for HS image classification; MambaHSI and GMamba are adapted with DSM streams matching their HS streams, and all methods are tuned under the common setup.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Results and Comparison

Experiments were conducted on the three datasets to demonstrate the effectiveness of ReportsNet. The classification accuracies for these datasets are presented in Tables IV–VI, respectively, with the best results highlighted in bold. Generally, pixel-wise techniques, that is, early, middle, late, and ENDnet, exhibit inferior performance due to their inability to comprehensively capture spatial information. Since the experimental results on the MUUFL and Trento datasets clearly verify the limitations of these pixel-wise techniques, they are not conducted on the Houston dataset.

Cross-modal methods generally outperform the single-modal CNN_H and CNN_L baselines, supporting HS-DSM complementarity on the evaluated datasets. The three tables also include RIC-CNN and RotEq-CNN as direct rotation-aware baselines. ReportsNet obtains the highest OAs of 91.99%, 99.45%, and 90.69% on MUUFL, Trento, and Houston, respectively. Its 0.44-point Trento margin over RIC-CNN is modest and is treated as numerical rather than practical superiority despite the repeated-run statistical result.

On MUUFL, the OA gains over HugIpuNet and CHNet are 1.44 and 1.48 points, respectively, and the resource comparison further contextualizes these gains. Trento is close to saturation, while on Houston ReportsNet improves OA by 1.28 points over the next-best method and obtains higher accuracies on difficult categories such as Roads and Cars.

Two-sided exact Wilcoxon signed-rank tests over ten matched runs compare ReportsNet with HugIpuNet on MUUFL and Houston and with RIC-CNN on Trento. The raw p-values are 0.003906, 0.001953, and 0.001953, respectively; after Holm correction, all adjusted values are 0.005859 and remain significant at $\alpha = 0.05$. These results establish statistically consistent OA differences under the evaluated protocol; practical relevance is assessed separately through effect magnitude, rotation stability, and resource use.

Local polar-radial resampling uses bilinear interpolation to map fractional sampling positions while retaining both source images on their registered grids. Results across the three datasets demonstrate the effectiveness of explicitly organizing context by radial distance and cyclic angle. Figs. 7–9 visualize the classification maps by different techniques on the MUUFL, Trento, and Houston datasets, respectively. ReportsNet generates classification maps that most closely align with the ground truth, exhibiting fewer misclassified samples and more homogeneous regions.

B. Performance Under Different Training Ratios

Table VII comprehensively evaluates unchanged architectures at 0.5%, 1%, and 3% class-wise training ratios.

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TABLE IV

CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), κ ($\times$100), PR (%), AND $F1$ ($\times$100), ON THE MUUFL DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	MSNetSC	LIIT	Sal^{2}RN	HDML	ExViT	MambaHSI	GMamba	HugIpuNet	CHNet	RIC-CNN	RotEq-CNN	Ours
1	94.42	94.36	92.64	93.75	96.57	84.27	96.20	96.72	95.28	97.21	96.52	97.56	97.29	97.21	**98.01**	97.90	97.03	97.57
2	51.40	64.10	51.49	71.14	63.55	33.96	74.01	66.91	73.89	69.37	71.33	71.26	77.37	74.79	73.31	**94.84**	76.34	82.54
3	82.03	65.60	74.08	66.74	76.12	35.40	81.08	79.14	80.41	77.78	80.55	81.37	81.39	83.55	80.10	54.48	79.63	**84.60**
4	69.03	73.34	80.92	47.51	59.02	4.48	91.54	75.88	88.44	77.71	83.46	77.16	84.57	85.29	86.01	**92.04**	86.50	87.78
5	93.23	86.72	91.10	88.94	93.04	62.81	92.66	91.47	94.64	90.95	90.20	**95.47**	93.70	94.59	92.45	94.88	90.82	92.78
6	72.89	83.73	73.97	82.00	45.55	74.40	95.66	89.15	85.03	63.12	1.74	48.81	51.41	**95.88**	90.89	85.03	80.91	93.06
7	47.04	64.18	62.96	67.62	62.60	15.65	81.28	69.74	77.02	75.35	82.32	64.63	69.52	77.97	**84.56**	80.51	77.34	83.58
8	86.73	90.84	88.94	91.19	95.03	76.17	97.17	95.60	95.40	96.62	97.90	97.65	96.65	97.38	96.03	97.26	94.92	**97.98**
9	57.91	50.40	61.05	47.78	41.58	2.41	34.21	32.53	33.77	45.73	21.88	41.36	43.03	57.18	63.75	60.98	32.97	**66.16**
10	24.86	28.73	25.41	15.47	0	0.55	0	25.41	0.55	3.87	1.10	14.92	0.56	0.66	8.84	1.10	0.00	**50.28**
11	41.73	59.40	56.39	**87.97**	66.92	0	40.98	45.11	59.02	31.58	50.00	62.03	20.68	43.98	77.44	66.92	54.51	48.12
OA	83.90	83.31	83.35	83.35	85.67	61.92	89.18	87.26	88.43	87.82	87.50	88.75	89.05	90.55	90.51	89.17	88.62	**91.99**
AA	65.57	69.22	69.00	69.10	63.63	35.46	71.34	69.79	71.22	66.30	61.54	68.38	65.11	74.04	77.40	75.09	70.09	**80.40**
κ	78.00	78.35	78.43	78.00	80.76	51.59	85.68	83.08	84.68	83.80	83.42	86.75	86.75	87.61	87.39	85.68	84.85	**89.39**
PR	75.19	68.96	73.87	68.61	72.55	37.56	71.65	72.95	77.88	81.11	75.87	79.43	80.53	87.61	80.70	**88.61**	77.11	84.99
F1	68.40	68.43	70.55	68.71	67.02	35.21	72.50	70.75	73.23	69.83	62.32	72.51	68.00	76.68	78.58	76.21	72.55	**82.29**

TABLE V

CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), κ ($\times$100), PR (%), AND $F1$ ($\times$100), ON THE TRENTO DATASET

Class	Early	Middle	Late	ENDnet	CNN_H	CNN_L	MSNetSC	LIIT	Sal^{2}RN	HDML	ExViT	MambaHSI	GMamba	HugIpuNet	CHNet	RIC-CNN	RotEq-CNN	Ours
1	66.85	64.37	82.37	69.23	99.10	89.26	92.21	96.19	98.60	98.97	99.50	99.57	99.57	98.40	97.50	99.87	98.25	99.17
2	97.63	95.09	97.22	95.06	89.98	79.75	97.77	94.33	94.89	91.75	97.70	97.43	98.40	97.56	98.78	**99.93**	92.66	97.43
3	91.14	93.88	86.29	78.69	94.73	13.71	90.08	98.10	86.92	92.83	93.25	98.59	80.17	99.37	69.83	94.94	91.98	97.05
4	99.25	98.77	99.07	98.48	99.35	92.36	98.49	99.79	99.97	99.86	**100.00**	99.97	99.88	**100.00**	**100.00**	**100.00**	99.98	**100.00**
5	90.04	91.84	88.27	87.17	99.84	98.02	96.50	99.90	98.86	98.94	**100.00**	97.04	97.67	99.44	**100.00**	**100.00**	99.95	**100.00**
6	89.88	90.80	87.62	91.15	77.59	76.89	92.17	97.58	93.41	96.94	93.95	97.36	98.12	96.59	93.06	91.57	**98.82**	98.57
OA	90.46	90.50	91.51	89.23	96.23	89.83	96.10	98.57	98.01	98.22	98.97	98.36	98.43	98.99	98.34	99.01	98.79	**99.45**
AA	89.13	89.13	90.14	86.63	93.43	75.00	94.54	97.65	95.44	96.55	97.40	98.38	95.64	98.56	93.20	97.72	96.94	**98.70**
κ	86.89	87.01	88.51	85.63	94.80	86.28	94.80	98.09	97.35	97.63	98.62	97.97	98.09	98.20	97.78	98.68	98.38	**99.26**
PR	88.77	90.52	88.71	88.16	91.74	81.87	92.81	95.68	97.23	97.25	98.98	97.97	98.09	98.20	98.14	97.83	**98.40**	98.37
F1	88.85	89.72	89.26	87.36	92.31	76.40	93.60	96.52	96.25	96.86	98.16	98.15	96.66	98.38	95.27	97.72	97.62	**98.53**

TABLE VI

CLASSIFICATION RESULTS, INCLUDING OA (%), AA (%), κ ($\times$100), PR (%), AND $F1$ ($\times$100), ON THE HOUSTON DATASET

Class	MSNetSC	LIIT	Sal^{2}RN	HDML	ExViT	MambaHSI	GMamba	HugIpuNet	CHNet	RIC-CNN	RotEq-CNN	Ours
1	78.39	50.73	71.16	**83.60**	78.79	80.82	78.04	73.59	73.78	55.35	76.63	77.75
2	77.03	74.96	89.44	83.82	79.80	81.99	84.62	86.67	**89.76**	87.15	77.69	81.97
3	85.73	86.38	95.21	96.88	96.58	97.39	97.49	**97.99**	93.14	94.46	95.48	96.58
4	82.43	84.68	83.86	87.69	88.99	86.80	79.25	85.13	88.99	84.85	90.26	**91.35**
5	76.35	**95.16**	86.22	91.73	89.54	86.34	86.76	91.29	89.23	86.12	91.96	95.08
6	93.27	93.89	96.81	97.01	97.10	96.85	96.03	98.00	97.54	96.28	97.61	**98.09**
7	36.65	44.08	46.85	44.64	37.22	45.99	42.85	54.64	42.63	50.58	25.51	**61.53**
8	68.63	58.97	56.98	71.86	64.80	70.15	69.55	**75.06**	68.19	68.57	62.53	71.52
9	72.23	78.15	89.74	86.60	88.26	84.84	87.25	85.46	88.03	**91.47**	86.75	90.49
10	86.63	89.57	89.67	91.61	**98.23**	93.32	91.98	92.32	95.62	97.47	94.34	93.00
11	29.27	54.40	51.38	28.26	76.33	63.39	63.30	79.17	76.33	65.50	81.56	**97.06**
12	94.46	96.05	98.08	96.48	99.73	99.22	96.46	97.99	99.88	**99.94**	99.60	99.78
OA	81.00	82.89	86.50	87.74	87.33	87.39	86.90	89.41	88.02	88.14	86.35	**90.69**
AA	73.42	75.59	79.62	80.02	82.71	82.26	81.13	84.78	83.18	81.48	81.66	**87.85**
κ	76.05	78.48	82.90	84.51	84.00	84.80	84.07	86.95	84.83	85.05	82.71	**88.24**
PR	75.03	78.54	81.22	85.24	83.96	84.80	84.07	86.95	85.24	86.35	83.99	**88.32**
F1	73.15	76.46	80.03	81.16	82.96	83.26	82.20	85.75	83.98	83.18	82.03	**88.04**

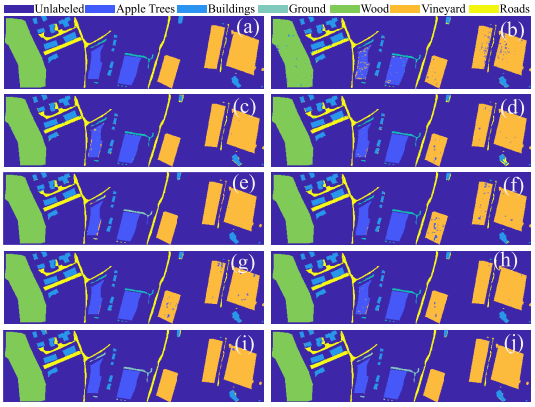


Fig. 8. (a) Ground truth and classification maps on the Trento dataset, including (b) MSNetSC, (c) LIIT, (d) HDML, (e) ExViT, (f) MambaHSI, (g) GMamba, (h) HugIpuNet, (i) CHNet, and (j) Ours.

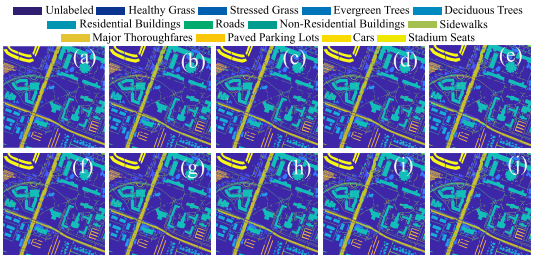


Fig. 9. (a) Ground truth and classification maps on the Houston dataset, including (b) MSNetSC, (c) LIIT, (d) HDML, (e) ExViT, (f) MambaHSI, (g) GMamba, (h) HugIpuNet, (i) CHNet, and (j) Ours.

TABLE VII

OA (%) UNDER DIFFERENT CLASS-WISE TRAINING RATIOS ON HOUSTON

Method	0.5%	1%	3%
CHNet	86.16	88.02	92.62
GMamba	85.04	86.90	91.14
MambaHSI	82.46	87.39	91.81
ExViT	85.94	87.33	91.54
HDML	84.89	87.74	90.63
Sal^{2}RN	84.72	86.50	91.55
HugIpuNet	86.99	89.41	92.55
RIC-CNN	85.22	88.14	92.15
RotEq-CNN	84.63	86.35	91.57
ReportsNet	**87.08**	**90.69**	**93.04**

ReportsNet ranks first at each ratio, with Houston OAs of 87.08%, 90.69%, and 93.04%.

Across the same ratios, ReportsNet obtains OAs of 89.25%, 91.99%, and 95.02% on MUUFL and 98.95%, 99.45%, and 99.63% on Trento, extending the limited-label evidence beyond the original 1% setting.

C. Spatially Independent Evaluation

For spatially independent Trento evaluation, 302 class-wise training centers from rows 155 to 165 and 27 877 test centers from rows 0 to 144 are separated by a ten-row guard band,

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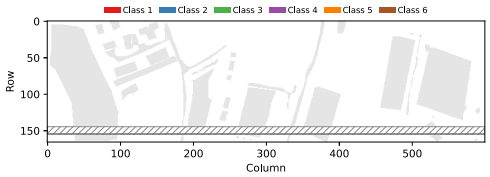


Fig. 10. Spatially independent training-sample locations on Trento. Gray pixels indicate the labeled scene footprint, colored pixels indicate the selected training centers of the six classes, and the hatched strip denotes the guard band.

TABLE VIII
COMPUTATIONAL RESOURCE COMPARISON ON A SINGLE 11×11 HOUSTON PATCH

Method	Params. (M)	FLOPs (M)	Infer. (ms)	Train (ms)
CHNet	23.462	254.724	22.845	115.234
GMamba	0.328	3.915	25.012	68.987
MambaHSI	0.122	0.837	15.234	71.892
ExViT	0.234	2.457	23.101	104.578
HDML	0.281	6.523	5.231	21.234
Sal^{2}RN	0.907	11.077	11.543	46.989
HugIpuNet	0.257	66.486	361.204	233.478
RIC-CNN	0.302	10.183	3.712	22.789
RotEq-CNN	0.017	99.510	13.012	87.234
ReportsNet	0.977	204.005	39.142	92.871

ensuring that their 11×11 patches share no pixels. Without a separate validation set, the fixed 200-epoch model is evaluated at the final epoch.

Fig. 10 shows the training locations; OA, AA, κ, PR, and $F1$ are 92.89%, 79.24%, 90.35%, 92.54%, and 82.58%, respectively. The 6.56-point OA decrease from the random split provides a more conservative estimate under spatial separation.

D. Computational Efficiency Analysis

All methods are profiled on one NVIDIA RTX 2080 Ti using an 11×11 Houston patch; inference uses batch size one, and two-sample training latency is divided per patch. PyTorch operator hooks and the TensorFlow graph profiler are used for FLOPs, so custom operators may cause small profiler-dependent differences.

Table VIII reports 0.977 M parameters, 204.005 M FLOPs, and 39.142/92.871 ms inference/training latency for ReportsNet. Its batch-one inference rate is approximately 26 patches/s; compared with CHNet, it uses 95.8% fewer parameters and 19.9% fewer FLOPs. Together with the lighter baselines in the table, these measurements characterize the accuracy-rotation-stability-cost tradeoff of ReportsNet.

E. Rotation Robustness Analysis

RE removes within-ring angular order, whereas RLE processes center-preserving rotation as cyclic line reordering using shared radial encoding, relative angular offsets, circular convolution, and subdirection averaging. This complementary design promotes stable representations under changes in local orientation. To evaluate this property, the held-out Houston test patches are rotated from 0° to 90° at 5° intervals, and the model is trained without rotation augmentation. All methods

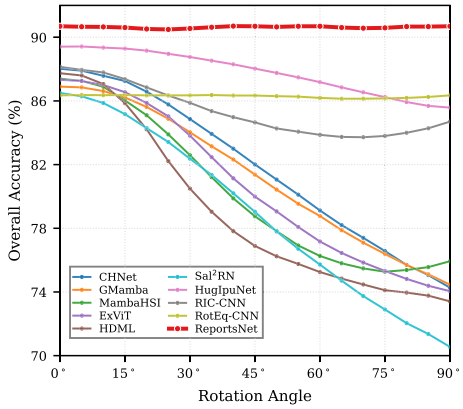


Fig. 11. OA curves of ReportsNet and nine representative baselines under rotations from 0° to 90° on the Houston dataset. All models are evaluated using the same rotation angles.

reuse their standard-comparison checkpoints and held-out test indices; thus, ReportsNet obtains 90.69% OA at 0° in both evaluations.

Fig. 11 shows that ReportsNet ranks first at every angle, with mean OA 90.63%, standard deviation 0.06%, and range 0.21 points. Baseline endpoint drops are 3.43–15.95 points, while RotEq-CNN has mean OA 86.32% and standard deviation 0.08%; RIC-CNN's angle dependence is consistent with pooling and interpolation effects noted in [22].

F. Ablation Study and Parameter Selection

1) Necessity of Both RS Modalities: To evaluate the contribution of the two inputs, several ReportsNet-based alternatives are constructed using HS imagery and the LiDAR-derived DSM as follows.

1) *HS+∅:* Only HS data are used, and the DSM branch is removed.
2) *DSM+∅:* Only the LiDAR-derived DSM is used, and the HS branch is removed.
3) *HS+HS:* Both inputs are HS data.
4) *DSM+DSM:* Both inputs are the DSM.
5) *HS+DSM (the proposed structure):* The inputs are one HS patch and one co-registered LiDAR-derived DSM patch.

Table IX shows that the DSM-only raster provides surface elevation but lacks spectral, return-intensity, point-density, and vertical-distribution information, making similar low-lying surfaces difficult to distinguish. HS+DSM improves OA over HS-only by 8.53, 2.78, and 6.54 points on MUUFL, Trento, and Houston, whereas duplicating either modality does not provide the same complementarity.

2) Necessity of Network Structure: S_1 /S_2 remove the RAM attention/convolution branch; S_3-S_5 remove radial encoding, use basic positional encoding, or remove all positional encoding. S_6-S_{10} remove RLE PCMA, RE, RE cross-attention, RLE, or SSA Mamba, respectively. S_{11} replaces RAM attention with $K \times K$ pairwise cross-line attention while retaining circular convolution; S_{12} is a capacity-matched Cartesian replacement,

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TABLE IX
MODALITY ABLATION RESULTS ON THE THREE DATASETS

Dataset	Scenario	OA (%)	AA (%)	κ ($\times$100)	PR (%)	F1 ($\times$100)
MUUFL	HS+$\varnothing$	83.46	64.83	77.92	74.85	68.06
	DSM+$\varnothing$	66.18	30.89	54.32	42.05	30.34
	HS+HS	89.61	76.15	86.26	83.45	78.79
	DSM+DSM	67.54	31.81	56.25	35.75	31.44
	HS+DSM	**91.99**	**80.40**	**89.39**	**84.99**	**82.29**
Trento	HS+$\varnothing$	96.67	93.45	95.56	93.00	93.20
	DSM+$\varnothing$	95.88	80.53	94.48	92.58	80.78
	HS+HS	97.56	95.74	96.74	95.86	95.75
	DSM+DSM	95.99	80.67	94.64	95.16	79.66
	HS+DSM	**99.45**	**98.70**	**99.26**	**98.37**	**98.53**
Houston	HS+$\varnothing$	84.15	77.12	79.81	81.26	78.77
	DSM+$\varnothing$	73.51	59.77	66.54	62.09	57.92
	HS+HS	88.34	83.66	85.29	84.74	84.06
	DSM+DSM	67.54	53.86	59.45	59.33	46.36
	HS+DSM	**90.69**	**87.85**	**88.24**	**88.32**	**88.04**

TABLE X
TRAINING AND TESTING SAMPLE NUMBERS IN THE MUUFL DATASET

Class	Land-cover Category	Training	Testing	Total
1	Trees	232	23014	23246
2	Mostly Grass	42	4228	4270
3	Mixed Ground Surface	69	6813	6882
4	Dirt and Sand	18	1808	1826
5	Road	67	6620	6687
6	Water	5	461	466
7	Building Shadow	22	2211	2233
8	Building	62	6178	6240
9	Sidewalk	14	1371	1385
10	Yellow Curb	2	181	183
11	Cloth Panels	3	266	269

S_{13} replaces PCMA with global attention, and S_{14} removes Sobel preprocessing.

Table X jointly reports classification performance and parameter count so that structural effects can be distinguished from capacity changes. Controlled attribution is provided by the S_{12} capacity-matched Cartesian control, S_{13} equal-parameter global-attention replacement, and S_{14} Sobel removal under matched experimental settings. Removing the RAM attention or convolution branch in S_1 and S_2 reduces OA by 0.76 and 1.89 points, respectively; removing PCMA in S_6 reduces OA by 1.47 points. Removing RE, its cross-attention pathway or RLE in S_7-S_9 reduces OA by 1.15, 0.87, and 2.26 points, while removing the SSA Mamba branch in S_{10} reduces OA by 0.73 points. Because gated fusion operates jointly with cross-attention, S_8 evaluates the RE interaction pathway rather than isolating an independent causal contribution of the gate.

The equal-capacity positional variants S_3-S_5 retain 0.977M parameters but lose 0.57–1.14 OA points. Conversely, the larger S_{11} and S_{12} controls contain 1.195M and 0.999M parameters but remain 2.18 and 2.20 points below ReportsNet, showing that capacity alone does not explain the result. Because S_{12} jointly replaces polar-radial resampling and its associated expert operators, its result supports the complete topology-aware design on Houston rather than isolating a causal advantage of coordinate conversion alone. S_{13} also retains 0.977M parameters but reduces OA/AA by 0.37/3.16 points and increases FLOPs from 204.005M to 251.089M while reducing inference latency from 39.142 to 38.283 ms and training latency from 92.871 to 92.447 ms under the same profiling protocol. It loses 0.16 OA points at 90° relative to 0°, whereas PCMA has equal endpoint OA. S_{14} retains the parameter count but reduces OA/AA by 0.35/2.80 points. Together, these controls distinguish the major topology, attention, and preprocessing contributions while treating the smaller variations as incremental effects within the interacting architecture.

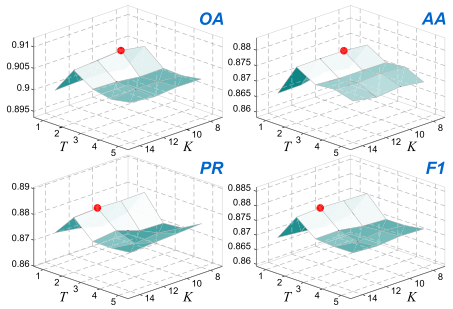


Fig. 12. Grid search results for the optimal K and T on the Houston dataset. The best result is highlighted.

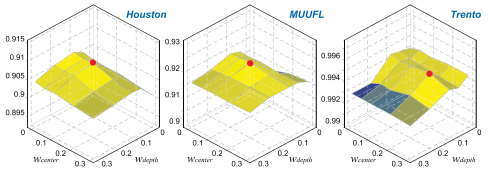


Fig. 13. Grid search results for the optimal w_{depth} and w_{center} on the three datasets. The best results are highlighted.

3) Architectural Parameter Selection: The radial line number K controls angular resolution, and RLE depth T controls capacity. The 20-point grid in Fig. 12 spans $K \in \{8, 10, 12, 14\}$ and $T \in \{1, \ldots, 5\}$, corresponding to 0.719M–1.752M parameters, 85.597M–697.371M FLOPs, 26.987–76.103 ms inference latency, and 78.012–131.456 ms training latency per patch. Validation selects $K = 10$ and $T = 3/5/2$ for MUUFL/Trento/Houston; larger models do not improve monotonically. A unified $K = 10, T = 2$ still obtains 90.39%, 99.30%, and 90.69% OA on MUUFL, Trento, and Houston, respectively.

4) Loss Weight Selection: $\mathcal{L}_{depth}$ and $\mathcal{L}_{center}$ supervise intermediate RLE and angular-order-pooled center predictions only during training. Fig. 13 searches both weights over $\{0, 0.05, 0.1, 0.2, 0.3\}$ and selects $w_{depth} = 0.1$ and $w_{center} = 0.1/0.2/0.1$ for MUUFL/Trento/Houston. This grid documents weight sensitivity but is not treated as an independent ablation proving either auxiliary loss.

VI. CONCLUSION

ReportsNet reorganizes target-centered Cartesian patches into complementary ring and radial-line tokens for distance-, relative-angle-, and geometry-aligned HS-DSM modeling. It obtains 91.99%, 99.45%, and 90.69% OA on MUUFL, Trento, and Houston, with the modest 0.44-point Trento margin reported as a numerical improvement. Houston rotation OA

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has a 0.06% standard deviation, the larger Cartesian control is 2.20 points lower, and unified $K = 10$, $T = 2$ retains 90.39%, 99.30%, and 90.69% OA. Resource measurements further establish its accuracy-rotation-stability-cost profile for joint HS and LiDAR-derived DSM classification. Future work will investigate richer LiDAR inputs, efficient operators, and broader RS tasks.

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